

Checking **payment certificates** with AI, as a product study.

Twelve stages, four on architecture. It advises, never signs.



Samy Yusif

AI Products Section Head

Nobody checks every line, so nobody knows the exposure.

Quantities are claimed against the bill and accumulate across certificates. Checking is done by sample because there is no time. The unchecked remainder is not zero risk, it is unmeasured risk.



Samy Yusif

AI Products Section Head

Sampling hides the number you actually need.

`exposure = unchecked x rate x avg error`

Value not checked, times the historical error rate, times the average size of an error. Pull the rate from certificates already closed and disputed. It exists, it has just never been added up.



Samy Yusif

AI Products Section Head

Rules first. The model only where the page is unstructured.

Reconciling cumulative quantities is arithmetic and belongs in a rules engine that cannot hallucinate. Bring the model in to read the messy document, never to decide the number on it.



Samy Yusif

AI Products Section Head

Checking moves from sampling to full coverage.

The engineer stops opening lines at random and starts working only the exceptions. Same person, same authority, different queue. Say this out loud early, because it sounds like a job threat.



Samy Yusif

AI Products Section Head

Subtract the cost of your own false alarms.

`cost = flags x minutes each x rate`

Value is errors recovered plus hours released. Against it sits the investigation cost of every flag that turns out fine. A tool that flags everything has a negative return and looks extremely busy.



Samy Yusif

AI Products Section Head

Extract, reconcile, escalate. The model touches only the first.

Read the certificate. Match every line to the bill and to the previous cumulative. Let the arithmetic settle whatever it can. Only what refuses to close reaches a human, carrying its page and line.



Samy Yusif

AI Products Section Head

Three quantities have to line up on every single item.

`bill qty, previous cum, this claim`

Keyed on the item code. And item codes drift between revisions of the bill, so that mapping is the genuinely hard part. Budget for it as work, not as a configuration step someone does on a Thursday.



Samy Yusif

AI Products Section Head

Arithmetic decides. The model never decides a number.

`settled by rules = never sent to model`

Any line the reconciliation can close is closed in code and never reaches a model at all. This is what caps your hallucination surface to the lines that were unreadable in the first place.



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AI Products Section Head

A flag with no page and line number does not ship.

Every exception points at where it came from, or the engineer spends longer verifying the flag than checking the line himself. That single rule decides whether the tool survives its second month.



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AI Products Section Head

A false negative costs money. A false positive costs the tool.

One missed overclaim leaves through the bank. A flood of wrong flags kills trust in week two and nobody opens it in week three. The two failures pull in opposite directions, so choose deliberately.



Samy Yusif

AI Products Section Head

The bill of quantities has to exist as data.

If it lives as a scanned PDF, that is the project you are actually funding, and it belongs in the plan and the budget instead of being discovered in week three of the pilot.



Samy Yusif

AI Products Section Head

Test on closed certificates where the errors are known.

```
recall  $\geq R$ , flags per cert  $\leq F$ 
```

Two thresholds, agreed before the run: how many of the known errors it must catch, and how many false flags per certificate the team will tolerate before they stop reading them at all.



Samy Yusif

AI Products Section Head

It advises. The engineer signs. That line never moves.

Automated checking of a formal contractual document changes nothing about who carries the signature. Put that in the rollout note, not in the small print of a slide nobody reads.



Samy Yusif

AI Products Section Head

Adoption is the share of certificates run before signature.

```
adoption = run before sign / total
```

Not logins, not reports opened. If certificates are still going out unchecked, the tool has not entered the process, whatever the usage dashboard is showing that month.



Samy Yusif

AI Products Section Head

Errors caught per certificate, against the sampling baseline.

`value = caught now - caught by sample`

You need the old number. If nobody recorded what sampling used to catch, take that baseline this month on the next certificate, before the tool touches it. Ten certificates beat zero.



Samy Yusif

AI Products Section Head

The hardest stage here is stage eight, and it is not AI.

Half the AI projects in contracting are data projects wearing a better title. Finding that out in stage eight is cheap. Finding it out in operations is not.



Samy Yusif

AI Products Section Head